

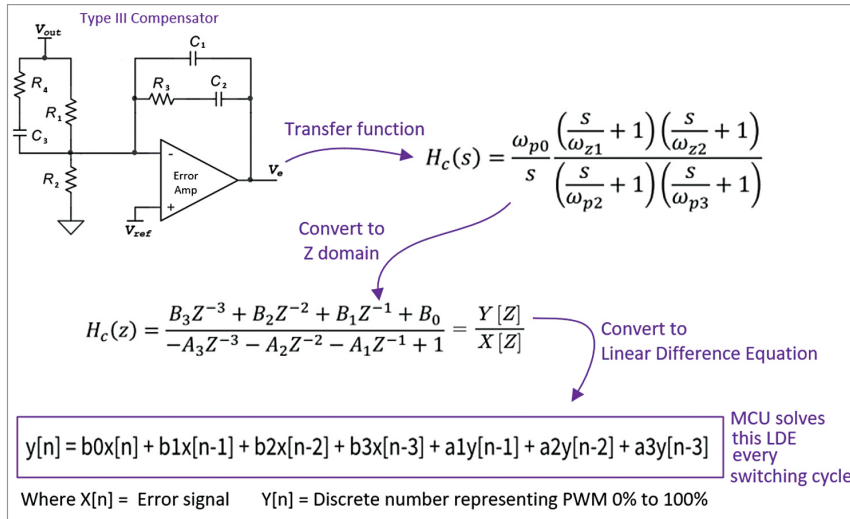


Fig. 8: The linear difference equation (LDE) replicating the mathematical operation of the op-amp analogue controller (Credit: Biricha Digital)

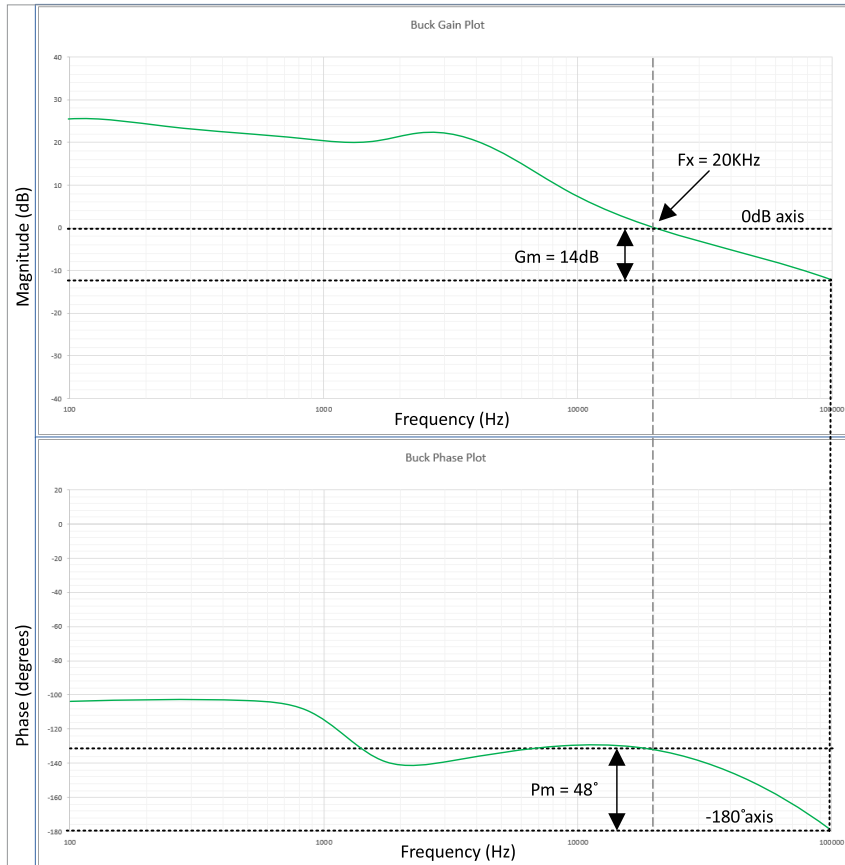


Fig. 9: Typical gain and phase plots of a buck regulator (Credit: www.infineon.com)

as the digital equivalent of an op-amp-based compensator. The linear difference equation (LDE) performs the same mathematical operations

as an op-amp. We need to convert the analogue controller/compensator transfer function $H_c(s)$ into a linear difference equation so that

MCU can calculate it. To convert the poles and zeros into the digital domain, the continuous time (s-domain) must be converted to the discrete (z-domain).

First, find $H(s)$ of a Type III/Type II compensator

Convert $H(s) \rightarrow H(z)$ then

Convert $H(z) \rightarrow$ LDE, which is what our MCU can solve for every switching cycle to generate a PWM

Coefficients $B_0, B_1, B_2, B_3, A_1, A_2, A_3$ are calculated and substituted into the LDE based on PSU design parameters, desired phase margin, crossover frequency, and poles and zeros placement, ensuring stability.

Stability criteria for switching regulators

Crossover frequency. This should be between 1/10 and 1/8 of the switching frequency.

Transient response. Higher crossover frequency results in faster transient response in the time domain

Phase margin. Must be $> 45^\circ$. A lower phase margin leads to oscillatory output

Gain margin. Should be $> 10\text{dB}$.

Gain plot slope. At crossover frequency (F_x), the slope should be -20dB/dec

Output capacitor ESR. Determines how shallow the gain falls at the crossover frequency, which should be -20dB/dec

Gain and phase crossing. Gain should cross 0dB before phase crosses -180° .

Application. Typical gain and phase plots are relevant for a buck regulator.

Future of digital power supply

As long as electronic devices exist, power consumption will remain a critical topic. With the growing emphasis on electrification for a cleaner environment and better efficiency, digital power supplies are poised to dominate, offering a promising future. **EFY**